2. ① $F = -2\rho g y S$ 简谐振动 $\omega = \sqrt{\frac{k}{m}} = \sqrt{\frac{2\rho g S}{m}}$ $(F = kx)$

② $E_p = (\rho g S y)\cdot y = \rho g S y^2$ 简谐振动 $\omega = \sqrt{\frac{k}{m}} = \sqrt{\frac{2\rho g S}{m}}$ $(E_p = \frac{1}{2}kx^2)$

$A\ddot{\eta} + B\eta = 0 \Rightarrow \omega = \sqrt{\frac{B}{A}}$ $A\dot{\eta}^2 + B\eta = const \Rightarrow \omega = \sqrt{\frac{B}{A}}$

3. 能量守恒 $E = \frac{1}{2}kx^2 + \frac{1}{2}mv^2 + \frac{1}{2}J\omega^2$. $J = \frac{1}{2}mR^2$. $v = R\omega$

$\Rightarrow E = \frac{1}{2}kx^2 + \frac{3}{4}mv^2 = const$ $E = \frac{1}{2}k x^2(t) + \frac{3}{4}m(x'(t))^2$

降次 对t求导 $\rightarrow kx\dot{x} + \frac{3}{2}m\dot{x}\ddot{x} = 0 \xRightarrow{约去\dot{x}} A = \frac{3}{2}m$. $B = k$ $\omega = \sqrt{\frac{B}{A}} = \sqrt{\frac{2k}{3m}}$.

4 1) 设 $\xi_1 = A_1\cos(\omega t + kx + \varphi)$. $\xrightarrow{x=-d} \xi_a = A_1\cos(\omega t - kd + \varphi) \Rightarrow \varphi = kd$

$\therefore \xi_1 = A_1\cos[\omega t + k(x+d)] = A_1\cos\omega(t - \frac{x+d}{u_1})$

2) $z = \rho u$ 媒质的"特性阻抗". 2大一定变

入射波在b处振动. $\xi_1(b) = A_1\cos\omega(t - \frac{l+d}{u_1})$

反射波: (半波损失) $\xi_1'(b) = A_1'\cos[\omega(t - \frac{l+d}{u_1}) - \pi]$

考虑b为振源. 到达坐标x需传播 $l-x$. $l-x$ 随t↑ 符号为 − (左加右减)

$\therefore \xi_1' = A_1'\cos[\omega(t - \frac{l+d}{u_1} - \frac{l-x}{u_1}) - \pi] = A_1'\cos[\omega(t + \frac{x-2l-d}{u_1}) - \pi]$

3) 透射波: $\xi = A\cos\omega(t - \frac{l+d}{u_1} - \frac{x-l}{u_2})$

同理 $\xi(c) = A\cos\omega(t - \frac{l+d}{u_1} - \frac{D}{u_2})$. 无半波损失. $\xi'(c) = \xi(c)$

$\therefore \xi_1'' = A_1''\cos\omega(t - \frac{l+d}{u_1} - \frac{D}{u_2} - \frac{D}{u_2} - \frac{l-x}{u_1}) = A_1''\cos\omega(t + \frac{x-2l-d}{u_1} - \frac{2D}{u_2})$

注意不同介质中波速不同

4) 相消干涉. 同一点相位差为π的奇数倍

$\omega(t + \frac{x-2l-d}{u_1}) - \pi - \omega(t + \frac{x-2l-d}{u_1} - \frac{2D}{u_2}) = (2k+1)\pi$ $k = 0, 1, 2 \cdots$

$\Rightarrow \frac{\omega D}{u_2} = (k+1)\pi$. $k = 0, 1, 2 \cdots$ $D = \frac{(k+1)\pi u_2}{\omega}$

k取0时. $D_{min} = \frac{\pi u_2}{\omega} = \frac{\pi u_2}{\frac{2\pi}{\lambda_2}} = \frac{\lambda_2}{2}$. "半透膜"

5. A: $y = A\cos\omega(t-\frac{x}{u})$ B: $y = A\cos[\omega(t-\frac{l-x}{u})-\pi]$

相位差 $\omega(t-\frac{x}{u})-\omega(t-\frac{l-x}{u})+\pi = (2k+1)\pi$, $k=0,1,2\cdots$

$\Rightarrow \omega\cdot\frac{l-2x}{u} = 2k\pi$ 代入数据 $\frac{\pi}{2}(30-2x) = 2k\pi \Rightarrow x = 15-2k$

考虑对称性. 静点至 $x = 1, 3, 5, 7, 9, 11, 13, 15, 17, 19, 21, 23, 25, 27, 29$.

法二. 驻波 $\lambda = 4m$ $l = 7.5\lambda$. A.B处波腹. 腹与节就差 $\frac{1}{4}\lambda = 1m$

节与节距离 $\frac{1}{2}\lambda = 2m$.

6. 拍频 $\nu = |\nu_1 - \nu_2|$ ($\omega_1 \approx \omega_2 \gg |\omega_1-\omega_2|$ 时形成"拍").

R测S, $\nu_1 = \frac{u-v_R}{u}\nu_S = \frac{270}{330}\nu_S = \frac{2700}{11}$ Hz

B测S. $\nu_1' = \frac{u-v_B}{u}\nu_S = \frac{230}{330}\nu_S = \frac{2300}{11}$ Hz

R测B: $\nu_2 = \frac{u+v_R}{u+v_B}\nu_1' = \frac{390}{430}\nu_1' = \frac{2300}{11}\times\frac{390}{430}$ Hz

$\nu = |\nu_1-\nu_2| = |\frac{2700}{11} - \frac{2300}{11}\times\frac{390}{430}| = 55.8$ Hz

1. (1) $\int_0^{v_0} av^2\,dv = \frac{1}{3}av_0^3 = 1 \Rightarrow a = \frac{3}{v_0^3}$

(2) $\bar{v} = \int_0^{+\infty} vf(v)\,dv = \int_0^{v_0} av^3\,dv = \frac{1}{4}av_0^4 = \frac{3}{4}v_0$.

(3) $\bar{v'} = \frac{\int_0^{v_0/2} vf(v)\,dv}{\int_0^{v_0/2} f(v)\,dv} = \frac{\int_0^{v_0/2} av^3\,dv}{\int_0^{v_0/2} av^2\,dv} = \frac{\frac{1}{64}av_0^4}{\frac{1}{24}av_0^3} = \frac{3}{8}v_0$.

2. 玻尔兹曼分布 $f(x) = Ae^{-\frac{E_p(x)}{kT}}$ $f(x)=n$. $A=n_0$; $f(x)=p$, $A=p_0$

r处离心势能 $E_p(r) = -\int_0^r m\omega^2 r\,dr = -\frac{1}{2}m\omega^2r^2$ (2分能关系)

$n = n_0e^{-\frac{E_p}{kT}} = n_0e^{\frac{m\omega^2r^2}{2kT}}$

3. $E_p(z) = mg(1-\frac{\rho_0}{\rho})z$ (等效重力势能) $n = n_0e^{-\frac{mg}{kT}(1-\frac{\rho_0}{\rho})z}$

4 (1) $f(v)dv = \begin{cases} 4\pi v^2dv\cdot D & . v\le v_F \\ 0 & . v>v_F \end{cases}$ 速度的立体角, x单位体积

$\therefore \int_0^{v_F} f(v)\,dv = \int_0^{v_F} \underbrace{4\pi v^2dv}_{体积}\underbrace{D}_{概率} = \frac{4}{3}\pi Dv_F^3 = 1$ $D = \frac{3}{4\pi v_F^3}$

$\therefore f(v) = \begin{cases} \frac{3v^2}{v_F^3} & . v\le v_F \\ 0 & . v>v_F \end{cases}$

(2) $\bar{v} = \int_0^{+\infty} v f(v)\,dv = \int_0^{v_F} \frac{3v^3}{v_F^3}\,dv = \frac{3}{4} v_F$

$f(v) = \begin{cases} \frac{3v^2}{v_F^3}, & v \le v_F \\ 0, & v > v_F \end{cases}$

$\sqrt{\overline{v^2}} = \sqrt{\int_0^{+\infty} v^2 f(v)\,dv} = \sqrt{\int_0^{v_F} \frac{3v^4}{v_F^3}\,dv} = \sqrt{\frac{3}{5}}\, v_F$

1. 绝热系统. 总$\Delta E = \Delta E_1 + \Delta E_2 = 0$. 即 $\nu_1 \cdot \frac{3}{2}R(T-T_1) + \nu_2 \frac{5}{2}R(T-T_2) = 0$

解得 $T = \frac{3\nu_1 T_1 + 5\nu_2 T_2}{3\nu_1 + 5\nu_2}$

$(\Delta T_1 = \Delta T_2)$ $(R = 8.31)$

2. (1) A. 等容. $\Delta Q_1 = \Delta E_1 = \frac{5}{2}R\Delta T$ B 等压 $\Delta Q_2 = C_p \Delta T = \frac{7}{2}R\Delta T$

$\Delta Q = 6R\Delta T = 335\,J \Rightarrow \Delta T = 6.72\,K$ $\Delta Q_1 = 139.61\,J$ $\Delta Q_2 = 195.45\,J$

(2). B. 等压绝热. $\Delta T_2 = 0$. $\Delta Q_2 = 0$ A. 等容 $\Delta Q_1 = \frac{7}{2}R\Delta T_1 = 335\,J$ $\Delta T_1 = 11.52\,K$

3. 绝热过程 $PV^\gamma = const$ $TV^{\gamma-1} = const$ $\frac{P^{\gamma-1}}{T^\gamma} = const$

(1). $PV^\gamma = const$. 全微分 $\Delta p V^\gamma + \gamma p V^{\gamma-1}\Delta V = 0$

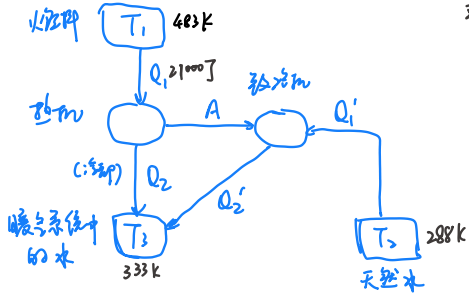
$\Rightarrow \Delta p = -\frac{\gamma p \Delta V}{V} = -\frac{\gamma p A}{V} \cdot y$ $\therefore F = \Delta p \cdot A = -\frac{\gamma p A^2}{V} y$

(2) $T = 2\pi\sqrt{\frac{mV}{\gamma p A^2}}$.

4. $\eta_0 = 1 - \frac{300}{1000} = 70\%$ $\eta_1 = 1 - \frac{300}{1100} = 72.73\%$ $\eta_2 = 1 - \frac{200}{1000} = 80\%$

可行性 提高高压蒸汽温度

5 动力暖气装置



1 kcal = 4.2J

求 $Q_2 + Q_2'$.

热机 $\eta = 1 - \frac{332}{483} = \frac{50}{161}$

$A = Q_1 \eta$

$Q_2 = Q_1(1-\eta)$

$Q_2' = A + Q_1' = A(1 + \frac{Q_1'}{A}) = A(1+w)$

$Q = Q_2 + Q_2' = Q_1 + Q_1' = Q_1 + Aw$

$= Q_1 + \eta Q_1 w$ $w = \frac{T_2}{T_3 - T_2}$

$= 62739.13\,kJ$

6. $Q_2 = Q_2' = \nu R T_2 \ln\frac{V_3}{V_4}$

$Q_2 = A\frac{T_2}{T_1-T_2} \Rightarrow Q_2' = Q_2 = A'\frac{T_2'}{T_1'-T_2'} = A\frac{T_2}{T_1-T_2} \Rightarrow T_1' = 425K$

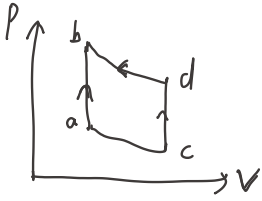
$\eta = 1-\frac{T_2'}{T_1'} = 29.4\%$

7. 绝热 $|W| = |\Delta E|$

	Q =	A +	ΔE
$1\to2\to3$	<0	>0	<0
$1\to2'\to3$	>0	>0	<0

看曲线的围面积，$|Q'| > |Q|$.

8.



$\Delta Q_{cd} = \Delta Q_{ab}$ 吸热

ΔQ_{db} 放热

$\therefore \Delta Q_2 < \Delta Q_1$

逐过程分析.

9. 可逆循环(卡诺循环)，输出功最大. 熵不变. 可逆绝热过程 $\Delta S = 0$

低温热源: $\Delta S_1 = \frac{Q_2}{T_0}$ $Q_2 \to$ 以放热

物体: $\Delta S_2 = \int_T^{T_0}\frac{dQ}{T} = \int_T^{T_0}\frac{CdT}{T} = C\ln\frac{T_0}{T}$

工质: $\Delta S_3 = 0$ (循环)

系统 $\Delta S = \Delta S_1 + \Delta S_2 + \Delta S_3 = C\ln\frac{T_0}{T} + \frac{Q_2}{T_0} = 0$

$W = Q_1 - Q_2 = C(T-T_0) + CT_0\ln\frac{T_0}{T}$

Q_1: 高温热源放热　Q_2: 低温热源吸热

1. (1) 真空. 自由膨胀 $W = Q = 0$

(2) 等温变化. $PV = const = P_0V_0$

$A = \int_{V_0}^{2V_0} PdV = \int_{V_0}^{2V_0}\frac{P_0V_0}{V}dV = P_0V_0\ln 2$　$\Delta E = 0$, $Q = A = P_0V_0\ln 2$.

(3) 不对.

2. $\Delta S = \nu C_V\ln\frac{T_2}{T_1} + \nu R\ln\frac{V_2}{V_1}$　$\Delta S = \nu C_V\ln\frac{P_2}{P_1} + \nu C_p\ln\frac{V_2}{V_1}$　$\Delta S = \nu C_p\ln\frac{T_2}{T_1} + \nu R\ln\frac{P_2}{P_1}$

等温自由膨胀 $\Delta S = R\ln\frac{V_2}{V_1}$

3. (1) 等压 同种 $V_1(V_2) \to \frac{V_1+V_2}{2}$　$\Delta S = \nu R\ln\frac{V_1+V_2}{2V_1} + \nu R\ln\frac{V_1+V_2}{2V_2} = \nu R\ln\frac{(V_1+V_2)^2}{4V_1V_2}$

(2) 不同种 $V_1(V_2) \to V_1+V_2$ (完散)　$\Delta S = \nu R\ln\frac{(V_1+V_2)^2}{V_1V_2}$

$(V \leftrightarrow P)$

4. (1) $2\times 312 + 1\times 300 = T$ $\quad T = 308\ K$

$$\Delta S = \int_{T_1}^{T}\frac{dQ_1}{T} + \int_{T_2}^{T}\frac{dQ_2}{T} = \int_{T_1}^{T}\frac{M_1 c\,dT}{T} + \int_{T_2}^{T}\frac{M_2 c\,dT}{T}$$

$$= M_1 c\ln\frac{T}{T_1} + M_2 c\ln\frac{T}{T_2} = 2.13\,J$$

注意c单位 $J\cdot g^{-1}\cdot K^{-1}$

(2) 玻尔兹曼熵公式 $S = k\ln\Omega$

$k = 1.38\times 10^{-23}$

开始状态 Ω_1，结束状态 Ω_2，平衡后又回到初始状态 状态 $\Omega = \frac{\Omega_1}{\Omega_2}$

$(\Omega_1 = \Omega_2\times\Omega)$

$$\Delta S' = k\ln\frac{\Omega_1}{\Omega_2} = -\Delta S \Rightarrow \frac{\Omega_1}{\Omega_2} = e^{-\frac{\Delta S}{k}} = 0$$

5. $T_0 = 263\ K$ $\quad T_1 = 273\ K$ $\quad T_2 = 288\ K$

$$\Delta S_{冰} = \int_{T_0}^{T_1}\frac{c_{冰}m\,dT}{T} + \frac{\lambda_{冰}m}{T_1} + \int_{T_1}^{T_2}\frac{c_{水}m\,dT}{T} = c_{冰}m\ln\frac{T_1}{T_0} + \frac{\lambda_{冰}m}{T_1} + c_{水}m\ln\frac{T_2}{T_1}$$

$$= 15.25\,J/K$$

$$\Delta S_{水} = -\frac{Q}{T_2} = -\frac{1}{T_2}\left(c_{冰}m(T_1 - T_0) + \lambda_{冰}m + c_{水}m(T_2 - T_1)\right) = -14.5\,J/K$$

$$\Delta S = \Delta S_{冰} + \Delta S_{水} = 0.75\,J/K.$$